

Course Outline for Computational Models in Cognitive Neuroscience (SFWRENG 4CN3)

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Teaching Assistants

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Lectures

Mon, Wed: 8:30-9:20 AM

Fri: 10:30 - 11:20 AM

PGCLL B131 (<https://library.mcmaster.ca/cct/class-dir/pgcll-b131>)

About the course

Understanding human brain activity holds the promise to solve a wide range of challenges, from creating brain-computer interfaces to understanding and addressing mental illnesses. However, despite intensive research, these aims remain elusive. One reason for this is the high complexity of the brain. Another big challenge is that in humans, measuring brain activity directly is highly invasive, and we thus rely largely on non-invasive, indirect measures. The goal of this course is giving you an introduction to Cognitive Neuroscience and a hands-on idea of what brain data looks like. We will focus on the Scientific Python ecosystem, which includes several connected open-source tools. By the end of this course, you will have an overview of commonly used measurement techniques, and hands-on experience using these measurements to answer questions about the brain. We will emphasize model-based data analysis, i.e., understanding how the computational tasks the brain performs relate to brain activity we can measure.

Learning Objectives

You will:

- Learn about common measures of human brain activity
- Be able to reason on the relationship between brain signals and neural processes
- Be able to formulate cognitive neuroscience questions as data analysis problems

- Be able to use open-source tools to analyze neural data
- Develop your data science skills
- Be able to critically evaluate information about the brain

Course Delivery

The course will consist of

- theoretical lectures following the textbook
- data analysis assignments
- methods lectures preparing you for specific components of the assignments

Readings will primarily be assigned textbook chapters, and may include additional primary sources.

Tentative list of theoretical topics:

- Ch. 2: Structure and Function of the Nervous System
- Ch. 3: Methods of Cognitive Neuroscience
- Ch. 4: Hemispheric Specialization
- Ch. 5: Sensation and Perception
- Ch. 11: Language
- Ch. 8: Action
- Ch. 9: Memory

Later topics may be changed based on student interests.

Tentative dates for assignments: Assignments will be given on Mondays, and are due on the Monday 14 days later. The topics of the data analysis assignments is provisional.

Assignment 1	Weeks 4-5	Basic manipulation of M/EEG and MRI data
Assignment 2	Weeks 6-7	Group analysis (event related potentials)
Assignment 3	Weeks 8-9	Temporal response function analysis
Assignment 4	Weeks 10-11	fMRI or iEEG

Resources

We will use the following **textbook** (\$72.95):

Gazzaniga, M. S., Ivry, R. B., & Mangun, G. R. (2019). *Cognitive neuroscience: The biology of the mind* (Fifth edition). W.W. Norton & Company.

You should have access to a **computer** with sufficient hard drive space (at least ~50 GB). This does not need to be on an internal drive, as data analysis can be performed using an external hard drive.

Assessments

You will work on 4 data analysis assignments, individually. There will be 2 quizzes (1 midterm quiz, and 1 at the end of term). The grade distribution is as follows:

Evaluation	Grade Weight	Topic
Assignment 1	15%	
Assignment 2	15%	
Assignment 3	15%	
Assignment 4	15%	
Midterm Quiz	20%	Lectures and readings Week 1 – 6
Final Quiz	20%	Cumulative, but focus on Weeks 7-12

- **Assignments:** These are structured data analysis projects. Each assignment has a dataset, and consists of a series of analysis tasks to be performed. Basic concepts will be discussed in class, but the assignments also require use of software documentation. The analyses are all based on real scientific questions, and should prompt you to think about underlying theoretical issues as well as getting the coding done. You will submit your code and text responses in Jupyter notebooks based on the instructions.
- **Mini-quizzes:** We will have several mini-quizzes to consolidate what we learn in the different textbook chapters. We will collectively decide in the first week whether these should count towards grades or not. If we decide they do, then they will collectively make up for 10% (which would reduce midterm and final weight to 15%).

Secondary sources

All available electronically. Can be consulted for extra information (will be references on slides).

Hedges, V. (2022). *Introduction to Neuroscience*. Michigan State University Libraries.

<https://openbooks.lib.msu.edu/introneuroscience1/>

Luck, S. J. (2014). *An introduction to the event-related potential technique* (Second edition). The MIT Press.

https://mcmaster.primo.exlibrisgroup.com/permalink/01OCUL_MU/deno1h/alma991032453667707371

Nunez, P. L., & Srinivasan, R. (2006). *Electric Fields of the Brain: The Neurophysics of EEG* (2nd ed.). Oxford University Press.
https://mcmaster.primo.exlibrisgroup.com/permalink/01OCUL_MU/6thkva/cdi_oup_oso_acprof_9780195050387

Requests for relief for missed academic term work

In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar “Requests for Relief for Missed Academic Term Work” (McMaster Student Absence Form (MSAF)). This class will adhere to the [University’s MSAF Guidelines](#).

- All missed Assignments will be granted a 3-day extension for submission.
- All missed Quizzes will have to be re-taken after the 3rd day of MSAF submission.

Resolving Issues

If the student believes they have been unfairly graded, they must access the Regrading Request Form posted on Avenue to report the issue. The form must be submitted via email to the instructor within 3 days of grade release. The instructor will respond to the Regrading Request within 3-5 business days, after receiving the form. Please do not complain or harass the TAs for your grades. Talk directly to the instructor.

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#), located at:

<https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one’s own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Plagiarism Detection

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty. Students who do not wish their work to be submitted through the plagiarism detection software must inform me before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details please refer to [Academic Integrity Policy](#).

Courses with online-element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student

disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

Equity, Diversity, and Inclusion

The Faculty of Engineering is committed to creating an environment in which students of all genders, cultures, ethnicities, races, sexual orientations, abilities, and socioeconomic backgrounds have equal access to education and are welcomed and treated fairly. If you have any concerns regarding inclusion in our Faculty, in particular if you or one of your peers is experiencing harassment or discrimination, you are encouraged to contact the Chair, Associate Undergraduate Chair, Academic Advisor or to contact the [Equity and Inclusion Office](#).

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

Academic Advising

For any academic inquiries please reach out to the Office of the Associate Dean (Academic) in Engineering located in JHE-Hatch 301.

Details on academic supports and contact information are available from:

<https://www.eng.mcmaster.ca/programs/academic-advising>

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.